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========================================
     SMART TRAFFIC MANAGEMENT SYSTEM
========================================
1. Vehicle Entry
2. Display Vehicles
3. Display Traffic Queues
4. Dequeue Vehicle
5. Traffic Density Analysis
6. Smart Signal Management
7. Emergency Vehicle Priority
8. Traffic Report
9. Shortest Path
10. Exit

Enter your choice: 1

Enter vehicle number: AP39CD5678
Enter vehicle type (Car/Bike/Bus/Ambulance/FireTruck): Ambulance
Enter road number (1-4): 2
Vehicle added successfully!
Emergency vehicle detected. Priority given.
```

```
Enter your choice: 1

Enter vehicle number: AP39AB1234
Enter vehicle type (Car/Bike/Bus/Ambulance/FireTruck): car
Enter road number (1-4): 1
Vehicle added successfully!
```

```
Enter your choice: 2

========== VEHICLE DETAILS ==========

Vehicle 1
Number  : AP39CD5678
Type    : Ambulance
Road    : 2
Priority: EMERGENCY

Vehicle 2
Number  : AP39AB1234
Type    : car
Road    : 1
Priority: NORMAL
```

```
Enter your choice: 3

========== TRAFFIC QUEUES ==========

Road 1:
AP39AB1234 - car

Road 2:
AP39CD5678 - Ambulance [EMERGENCY]

Road 3:
No vehicles

Road 4:
No vehicles
```

```
Enter your choice: 6

========== SMART SIGNAL ==========

1. Road 1
   Vehicles: 1
   Green Light: 20 seconds

2. Road 2
   Vehicles: 1
   Green Light: 20 seconds

3. Road 3
   Vehicles: 0
   Green Light: 20 seconds

4. Road 4
   Vehicles: 0
   Green Light: 20 seconds
```

```
Enter your choice: 7

========== EMERGENCY PRIORITY ==========
No emergency vehicles currently waiting.
```

```
Enter your choice: 8

========== TRAFFIC REPORT ==========
Total vehicles entered: 2
Vehicles on Road 1: 1
Vehicles on Road 2: 1
Vehicles on Road 3: 0
Vehicles on Road 4: 0
Highest-density road: Road 1
Emergency vehicles waiting: 0

Signal Timings:
Road 1: 20 seconds
Road 2: 20 seconds
Road 3: 20 seconds
Road 4: 20 seconds
```

```
Enter your choice: 9

Enter source junction (1-4): 2
Enter destination junction (1-4): 4

Shortest distance: 3 units
Shortest Path: 2 -> 4
```

```
Enter your choice: 4

Enter road number: 2
Emergency vehicle AP39CD5678 passed first from Road 2.
```

```
Enter your choice: 5

========== TRAFFIC DENSITY ==========
Road 1: 1 vehicles - LOW
Road 2: 0 vehicles - LOW
Road 3: 0 vehicles - LOW
Road 4: 0 vehicles - LOW
```

```
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Enter your choice: 10
Exiting Smart Traffic Management System.
```